Key features of parabolas

1 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$.

a If $a < 0$, in what direction will the parabola open?

b If $a > 0$, in what direction will the parabola open?

2 Does the parabola represented by the equation $y = x^2 - 8x + 9$ open upward or downward?

3 Does the graph of $y = x^2 + 6$ have any x-intercepts? Explain your answer.

4 State whether the following parabolas have x-intercepts:

a $y = (x-7)^2 + 4$

b $y = -(x-7)^2 + 4$

c $y = -(x-7)^2 - 4$

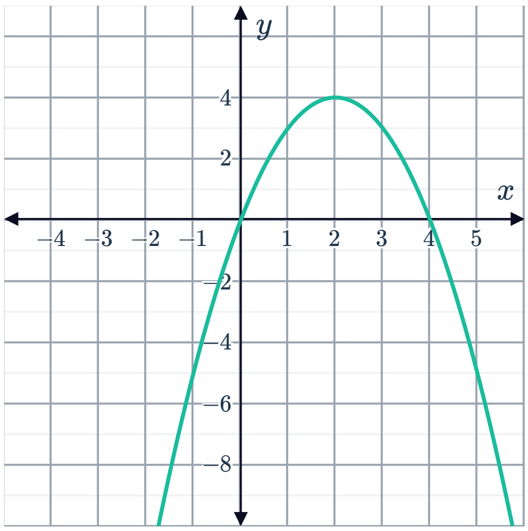
d $y = (x-7)^2 - 4$

5 Consider the given graph:

a What are the x-intercepts?

b What is the y-intercept?

c What is the maximum value?



6 Consider the given graph:

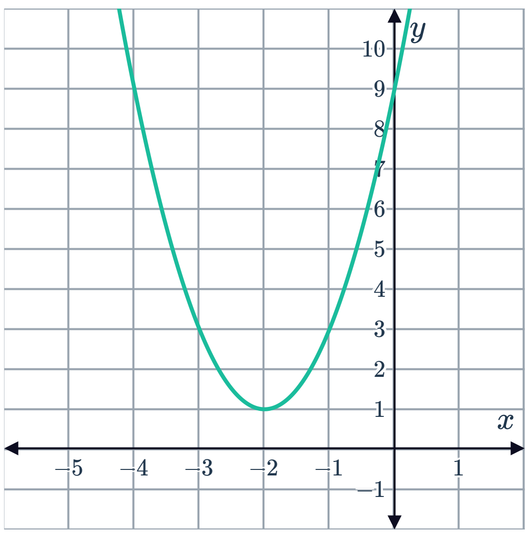
a Is the curve concave up or concave down?

b State the y-intercept of the graph.

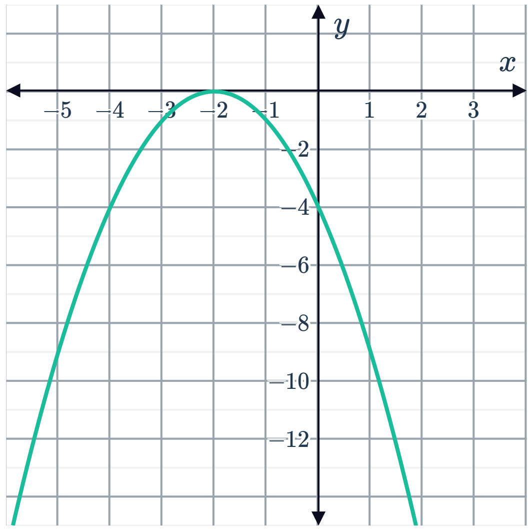
c What is the minimum value?

d At which value of x does the minimum value occur?

e Determine the interval of x for which the graph is decreasing.



7 Consider the graph of the parabola:



a State the coordinates of the x-intercept.

b State the coordinates of the vertex.

c State whether the following statements are true about the vertex:

i The vertex is the minimum value of the graph.

ii The vertex occurs at the x-intercept.

iii The vertex lies on the axis of symmetry.

iv The vertex is the maximum value of the graph.

8 Suppose that a particular parabola is concave down, and its vertex is located in quadrant 2.

a How many x-intercepts will the parabola have?

b How many y-intercepts will the parabola have?

9 Suppose that a particular parabola has two x-intercepts, and its vertex is located in quadrant 4. Will such a parabola be concave up or concave down?

10 Consider the quadratic function defined in the table on the right:

x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

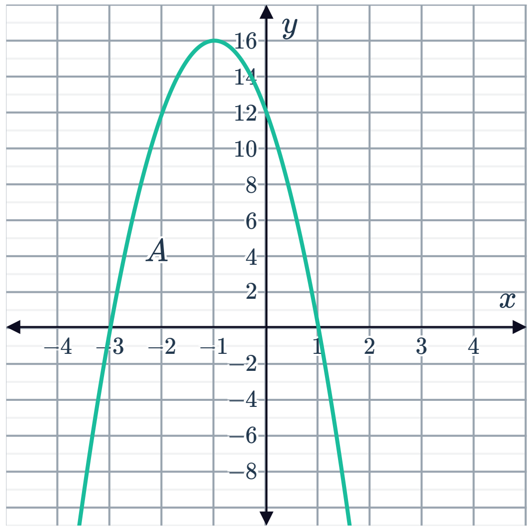
a What are the coordinates of the vertex?

b What is the minimum value of the function?

11 A vertical parabola has an x-intercept at $(-1, 0)$ and a vertex at $(1, -6)$. Find the other x-intercept.

12 State whether the following can be found, without any calculation, from the equation of the form $y = (x-h)^2 + k$ but not from the equation of the form $y = x^2 + bx + c$:

13 Quadratic function A is represented graphically as shown. Quadratic function B, which is concave down, shares the same x-intercepts as quadratic function A, but has a y-intercept closer to the origin. Which of the functions has a greater maximum value?



14 What is the axis of symmetry of the parabola $y = k(x-7)(x+7)$ for any value of k?

15 Consider the equation $y = 25 - (x+2)^2$. What is the maximum value of y?

16 Consider the function $y = (14-x)(x-6)$.

a State the zeros of the function.

b Find the axis of symmetry.

c Is the graph of the function concave up or concave down?

d Determine the maximum y-value of the function.

17 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.

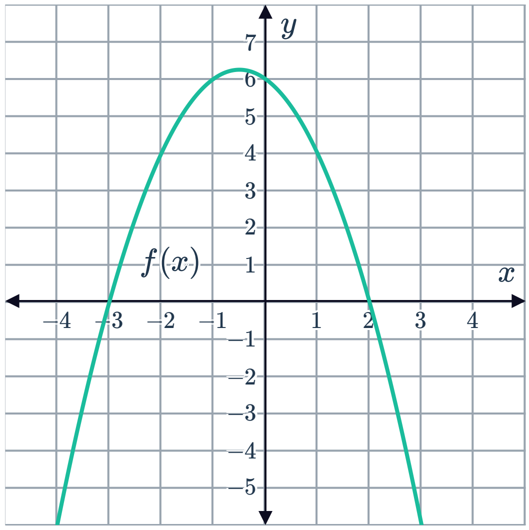
Complete the following statement:

The x-coordinate of the vertex of the parabola occurs at $x = \square$. The y-coordinate of the vertex is found by substituting $x = \square$ into the parabola's equation and evaluating the function at this value of x.

18 Consider the graph of the function

$f(x) = -x^2 - x + 6$:

Using the graph, write down the solutions of the equation $-x^2 - x + 6 = 0$.



19 True or false:

- **a** The quadratic formula can be used to find the y-intercept.
- **b** If the parabola has only one x-intercept , then the x-intercept is also the vertex.

20 Consider the parabola whose equation is $y = 3x^2 + 3x - 7$. Find the x-intercepts of the parabola in exact form.

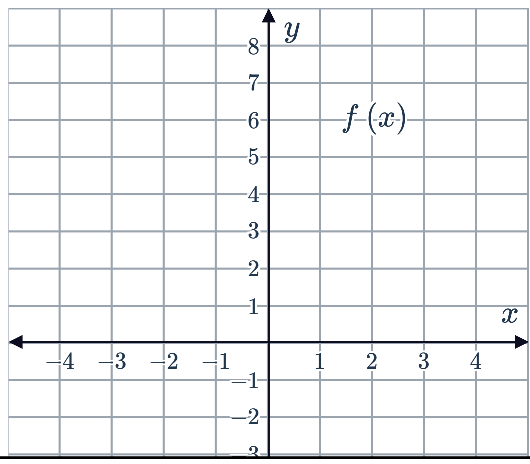
Graphing quadratics

21 Consider the parabola described by the function $y = -2x^2 + 2$.

- **a** Is the parabola concave up or down?
- **b** Is the parabola more or less steep than the parabola $y = x^2$?
- **c** What are the coordinates of the vertex of the parabola?
- **d** Sketch the graph of $y = -2x^2 + 2$.

22 Consider the two graphs. One of them has equation $f(x) = -x^2 + 5$.

- **a** What is the equation of the other graph?
- **b** Use the graph to approximate the solutions of the equation $-x^2 + 5 = 0$ to one decimal place.



23 Consider the quadratic function $h(x) = x^2 + 2$.

- **a** Sketch the graph of the parabola $h(x)$.
- **b** Plot the axis of symmetry of the parabola on the same graph.
- **c** What is the vertex of the parabola?

24 Consider the equation $y = (x-3)^2 - 1$.

- **a** Find the x-intercepts.
- **b** Find the y-intercept.
- **c** Determine the coordinates of the vertex.
- **d** Sketch the graph.
- **e** Use your graph to find the solutions of the equation $(x-3)^2 - 1 = 0$.

25 Consider the quadratic function $f(x) = -3(x+2)^2 - 4$.

- **a** What are the coordinates of the vertex of this parabola?
- **b** What is the equation of the axis of symmetry of this parabola?
- **c** What is the y-coordinate of the graph of $f(x)$ at $x = -1$?
- **d** Sketch the graph of the parabola.
- **e** Plot the axis of symmetry of the parabola on the same graph.

26 On a number plane, sketch the shape of a parabola of the form $y = a(x-h)^2 + k$ that has the following signs for a, h and k:

- **a** $a > 0, h > 0, k > 0$
- **b** $a < 0, h > 0, k > 0$
- **c** $a > 0, h > 0, k < 0$
- **d** $a < 0, h > 0, k < 0$

27 Consider the parabola $y=(2-x)(x+4)$.

a State the y-intercept.

b State the x-intercepts.

c Complete the table of values:

x	-5	-3	-1	1	3
y					

d Determine the coordinates of the vertex of the parabola.

e Sketch the graph of the parabola.

28 Consider the parabola $y=(x-3)(x-1)$.

a Find the y-intercept.

b Find the x-intercepts.

c State the equation of the axis of symmetry.

d Find the coordinates of the turning point.

e Sketch the graph of the parabola.

29 Consider the parabola $y=x(x+6)$.

a Find the y-intercept.

b Find the x-intercepts.

c State the equation of the axis of symmetry.

d Find the coordinates of the turning point.

e Sketch the graph of the parabola.

30 Sketch the graph of the following:

a $y=(x+2)(x-3)$

b $y=(x-3)(x+1)$

31 Consider the function $y=(x+5)(x+1)$.

a Sketch the graph.

b Sketch the graph of $y=-(x+5)(x+1)$ on the same set of axes.

32 Consider the equation $y=x^2-6x+8$.

a Factorise the expression x^2-6x+8.

b Hence, or otherwise, find the x-intercepts of the quadratic function $y=x^2-6x+8$

c Find the coordinates of the turning point.

d Sketch the graph of the function.

33 Consider the parabola $y = x^2 + x - 12$.

a Find the x-intercepts of the curve.

b Find the y-intercept of the curve.

c What is the equation of the vertical axis of symmetry for the parabola?

d Find the coordinates of the vertex of the parabola.

e Sketch the graph of $y = x^2 + x - 12$.

34 A parabola has the equation $y = x^2 + 4x - 1$.

a Express the equation of the parabola in the form $y = (x - h)^2 + k$ by completing the square.

b Find the y-intercept of the parabola.

c Find the vertex of the parabola.

d Is the parabola concave up or down?

e Hence, sketch the graph of $y = x^2 + 4x - 1$.

f Use your graph to approximate the solutions of the equation $x^2 + 4x - 1 = 0$.

35 Consider the quadratic $y = x^2 - 12x + 32$.

a Find the zeros of the quadratic function.

b Express the equation in the form $y = a(x - h)^2 + k$ by completing the square.

c Find the coordinates of the vertex of the parabola.

d Hence, sketch the graph.

36 Consider the curve $y = x^2 + 6x + 4$.

a Determine the axis of symmetry.

b Hence, determine the minimum value of y.

c Sketch the graph of the function.

d Use your graph to approximate the solutions of the equation $x^2 + 6x + 4 = 0$.

37 Consider the function $P(x) = -2x^2 - 8x + 2$.

a Find the coordinates of the vertex.

b Sketch the graph.

c Use your graph to approximate the solutions of the equation $-2x^2 - 8x + 2 = 0$

38 Consider the equation $y=6x-x^2$.

a Find the x-intercepts of the quadratic function.

b Find the coordinates of the turning point.

c Sketch the graph.

39 A parabola is described by the function $y=2x^2+9x+9$.

a Find the x-intercepts of the parabola.

b Find the y-intercept for this curve.

c Find the axis of symmetry.

d Find the y-coordinate of the vertex of the parabola.

e Sketch the graph.

Graphing quadratics using technology

40 Use your calculator or other handheld technology to graph the equations below. Then answer the following questions:

i What is the vertex of the graph?

ii What is the y-intercept?

a $y=4x^2-64x+263$

b $y=-4x^2-48x-140$

41 Use your calculator or other handheld technology to graph $y=-3x^2-12$.

a What is the vertex of the graph?

b Are there any x-intercepts?

c For what values of x is the parabola decreasing?

42 Use technology to graph the parabola $y=-2x^2+16x-24$.

a Find the x-intercepts of the parabola.

b Find the y-intercept of the parabola.

c Find the axis of symmetry of the parabola.

d Find the y-coordinate of the vertex of the parabola.